

Hyperbolic Functions

Define

$$\sinh x = \text{sh } x = \frac{e^x - e^{-x}}{2}$$

$$\cosh x = \text{ch } x = \frac{e^x + e^{-x}}{2}$$

$$\tanh x = \text{th } x = \frac{\text{sh } x}{\text{ch } x} = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

$$\coth x = \frac{\text{ch } x}{\text{sh } x} = \frac{e^x + e^{-x}}{e^x - e^{-x}}$$

$$\text{sech } x = \frac{1}{\text{ch } x} = \frac{2}{e^x + e^{-x}}$$

$$\text{cosech } x = \frac{1}{\text{sh } x} = \frac{2}{e^x - e^{-x}}$$

Remark:

As $\sin^2 x + \cos^2 x = 1$ we find $\text{ch}^2 x - \text{sh}^2 x = 1$

$\frac{d}{dx} \sin x = \cos x \longrightarrow \frac{d}{dx} \text{sh } x = \text{ch } x$

$\sin(x+y) = \sin x \cos y + \cos x \sin y \longrightarrow$ we find $\text{sh}(x+y) = \text{sh } x \text{ch } y + \text{ch } x \text{sh } y$

Example: prove that

i) $\cosh^2 x - \sinh^2 x = 1$

ii) $\sinh(x+y) = \sinh x \cosh y + \cosh x \sinh y$

Solⁿ:

i) LHS = $\cosh^2 x - \sinh^2 x$

$$= \left(\frac{e^x + e^{-x}}{2} \right)^2 - \left(\frac{e^x - e^{-x}}{2} \right)^2$$

$$= \frac{(e^x + e^{-x})^2}{4} - \frac{(e^x - e^{-x})^2}{4}$$

$$= \frac{e^{2x} + e^{-2x} + 2}{4} - \frac{e^{2x} + e^{-2x} - 2}{4}$$

$$= \frac{e^{2x} + e^{-2x} + 2 - e^{2x} - e^{-2x} + 4}{4}$$

$$= \frac{4}{4} = 1 = \text{RHS.}$$

$$\text{ii) } RHS = \sinh x \cosh y + \cosh x \sinh y$$

$$= \left(\frac{e^x - e^{-x}}{2} \right) \left(\frac{e^y + e^{-y}}{2} \right) + \left(\frac{e^x + e^{-x}}{2} \right) \left(\frac{e^y - e^{-y}}{2} \right)$$

$$= \frac{e^{x+y} + e^{x-y} + e^{-x+y} + e^{-(x+y)}}{4} + \frac{e^{x+y} - e^{x-y} - e^{-x+y} + e^{-(x+y)}}{4}$$

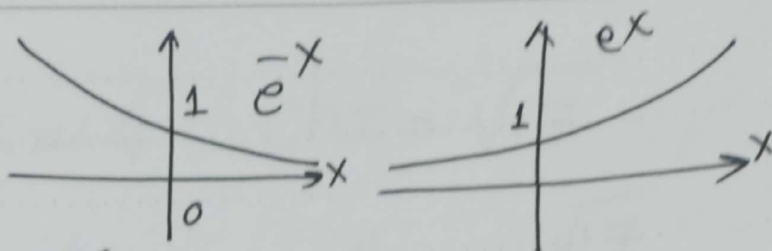
$$= \frac{e^{x+y} + \cancel{e^{x-y}} + \cancel{e^{-x+y}} + e^{-(x+y)}}{4} + \frac{e^{x+y} - \cancel{e^{x-y}} - \cancel{e^{-x+y}} + e^{-(x+y)}}{4}$$

$$= \frac{2e^{x+y} - 2e^{-(x+y)}}{4}$$

$$= \frac{e^{x+y} - e^{-(x+y)}}{2}$$

$$= \sinh(x+y) = LHS.$$

Remark:



$$\cosh x = \frac{e^x + e^{-x}}{2} > 0 \text{ for all } x$$

$$\sinh x = \frac{e^x - e^{-x}}{2} \begin{cases} > 0 \text{ if } x > 0 \\ < 0 \text{ if } x < 0 \end{cases}$$

Example: Given $\sinh x = 4$ find all other hyperbolic values of x .

Solution: $\cosh^2 x - \sinh^2 x = 1$

$$\Rightarrow \cosh^2 x = 1 + \sinh^2 x = 1 + 16 = 17$$

$$\Rightarrow \cosh^2 x = 17 \Rightarrow \cosh x = \pm \sqrt{17}$$

however, $\cosh x = -\sqrt{17}$ is rejected

$$\Rightarrow \boxed{\cosh x = \sqrt{17}}$$

ution:

Now, $\operatorname{sh} x = 4$ $\operatorname{ch} x = \sqrt{17}$

then $\operatorname{th} x = \frac{4}{\sqrt{17}}$ $\operatorname{coth} x = \frac{\sqrt{17}}{4}$

$\operatorname{sech} x = \frac{1}{\sqrt{17}}$ $\operatorname{cosech} x = \frac{1}{4}$

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Example: Repeat the above example replacing $\operatorname{ch} x = 4$

solⁿ: $\operatorname{ch} x = 4$

$\operatorname{sh}^2 x = \operatorname{ch}^2 x - 1 = 16 - 1 = 15$

$\Rightarrow \operatorname{sh} x = \pm \sqrt{15}$ (both accepted)

$\operatorname{sh} x$
 $\sqrt{15}$ OR $-\sqrt{15}$

$\operatorname{ch} x = 4$

$\operatorname{th} x = \sqrt{15}/4$

$\operatorname{coth} x = 4/\sqrt{15}$

$\operatorname{sech} x = \frac{1}{4}$, $\operatorname{cosech} x = \frac{1}{\sqrt{15}}$

$\operatorname{ch} x = 4 \Rightarrow \operatorname{th} x = -\frac{\sqrt{15}}{4}$

$\operatorname{coth} x = -4/\sqrt{15}$

$\operatorname{sech} x = \frac{1}{4}$

$\operatorname{cosech} x = -1/\sqrt{15}$

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Example: Prove that

i) $1 - \tanh^2 x = \operatorname{sech}^2 x$

ii) $\coth^2 x - 1 = \operatorname{cosech}^2 x$

Solⁿ: (i) $\cosh^2 x - \sinh^2 x = 1$ □

multiply both sides by $\frac{1}{\cosh^2 x}$ then

$$\frac{\cosh^2 x}{\cosh^2 x} - \frac{\sinh^2 x}{\cosh^2 x} = \frac{1}{\cosh^2 x}$$

$$\Rightarrow 1 - \tanh^2 x = \operatorname{sech}^2 x \quad \checkmark$$

(ii) multiply both sides of □ by $\frac{1}{\sinh^2 x}$

then $\frac{\cosh^2 x}{\sinh^2 x} - \frac{\sinh^2 x}{\sinh^2 x} = \frac{1}{\sinh^2 x}$

$$\rightarrow \coth^2 x - 1 = \operatorname{cosech}^2 x \quad \checkmark$$

Example: given $\text{th } x = \frac{1}{2}$, find all other hyperbolic values of x .

Solⁿ:

$$1 - \text{th}^2 x = \text{sech}^2 x$$

$$\Rightarrow \text{sech}^2 x = 1 - \left(\frac{1}{2}\right)^2 = \frac{3}{4}$$

$$\Rightarrow \text{sech } x = \pm \frac{\sqrt{3}}{2} \quad (-\text{ve is rejected})$$

$$\Rightarrow \boxed{\text{sech } x = + \frac{\sqrt{3}}{2}}$$

$$\Rightarrow \boxed{\cosh x = \frac{2}{\sqrt{3}}}$$

$$\text{th } x = \frac{\text{sh } x}{\text{ch } x} \Rightarrow \frac{1}{2} = \frac{\text{sh } x}{2/\sqrt{3}}$$

$$\text{then } \boxed{\text{sh } x = \frac{1}{\sqrt{3}}}, \boxed{\text{cosech } x = \sqrt{3}}$$

$$\boxed{\coth x} = \frac{1}{\text{th } x} = \boxed{2}$$

Exercise: Given $\operatorname{cosech} x = 3$

, find all other hyperbolic values.
(Do it yourself)

Derivatives of hyperbolic fns.

$$\frac{d}{dx} \operatorname{sh} x = \operatorname{ch} x \quad \frac{d}{dx} \operatorname{ch} x = \operatorname{sh} x$$

$$\frac{d}{dx} \operatorname{th} x = \operatorname{sech}^2 x \quad \frac{d}{dx} \operatorname{coth} x = -\operatorname{cosech}^2 x$$

$$\frac{d}{dx} \operatorname{sech} x = -\operatorname{sech} x \operatorname{th} x$$

$$\frac{d}{dx} \operatorname{cosech} x = -\operatorname{cosech} x \operatorname{coth} x$$

Proof: $\circledast y = \operatorname{sh} x = \frac{e^x - e^{-x}}{2}$

$$= \frac{1}{2} (e^x - e^{-x})$$

$$\Rightarrow y' = \frac{1}{2} (e^x - (-1)e^{-x})$$

$$y' = \frac{1}{2}(e^x + e^{-x}) = \cosh x$$

So $\frac{d}{dx} \sinh x = \cosh x$

Similarly, you can show that

$$\frac{d}{dx} \cosh x = \sinh x$$

$$* y = \tanh x = \frac{\sinh x}{\cosh x}$$

$$y' = \frac{\cosh x \cosh x - \sinh x \sinh x}{\cosh^2 x}$$

$$= \frac{\cosh^2 x - \sinh^2 x}{\cosh^2 x} = \frac{1}{\cosh^2 x} = \operatorname{sech}^2 x$$

Similarly, you can show that

$$\frac{d}{dx} \coth x = -\operatorname{cosech}^2 x$$

$$\frac{d}{dx} \operatorname{sech} x = -\operatorname{sech} x \tanh x$$

$$* y = \operatorname{sech} x = \frac{1}{\cosh x}$$

$$y' = \frac{0 - \sinh x}{\cosh^2 x}$$

$$= -\frac{\sinh x}{\cosh x} \cdot \frac{1}{\cosh x}$$

$$= -\tanh x \operatorname{sech} x$$

Similarly, you can show that

$$\frac{d}{dx} \operatorname{cosech} x = -\operatorname{cosech} x \coth x$$

Examples: Find $y'(x)$ for:

(a) $y = \cosh(x^3 + e^{-x})$ (b) $y = (\tanh(\ln x))^4$

(c) $y = \frac{\sinh 4x}{1 + \cosh 2x}$ (d) $y = \sqrt{\operatorname{sech}(\frac{1}{x})}$

(e) $y = e^{-x^2} \coth \sqrt{x}$ (f) $y = \ln(\operatorname{cosech} 4x)$

Solution:

(a) $y = \operatorname{ch}(x^3 + e^{-x})$

$$y' = \operatorname{sh}(x^3 + e^{-x}) \cdot (3x^2 - e^{-x})$$

(b) $y = (\operatorname{th}(\ln x))^4$

$$y' = 4(\operatorname{th}(\ln x))^3 (\operatorname{sech}^2(\ln x)) \frac{1}{x}$$

(c) $y = \frac{\operatorname{sh} 4x}{1 + \operatorname{ch} 2x} \rightarrow y' = \frac{4\operatorname{ch} 4x(1 + \operatorname{ch} 2x) - 2\operatorname{sh} 2x \operatorname{sh} 4x}{(1 + \operatorname{ch} 2x)^2}$

(d) $y = \sqrt{\operatorname{sech}(\frac{1}{x})} = (\operatorname{sech}(\frac{1}{x}))^{1/2}$

$$y' = \frac{1}{2} (\operatorname{sech}(\frac{1}{x}))^{-1/2} (-\operatorname{sech} \frac{1}{x} \operatorname{th} \frac{1}{x}) (-\frac{1}{x^2})$$

(e) $y = e^{-x} \operatorname{coth} \sqrt{x}$

$$y' = -e^{-x} \operatorname{coth} \sqrt{x} + e^{-x} (-\operatorname{sech}^2 \sqrt{x}) (\frac{1}{2\sqrt{x}})$$

$$f) y = \ln(\operatorname{cosech} 4x)$$

$$y' = \frac{1}{\operatorname{cosech}(4x)} (-\operatorname{cosech}(4x) \coth(4x) (4))$$

Inverse hyperbolic functions:

$$y = \sinh^{-1} x \rightarrow x = \sinh y$$

$$y = \cosh^{-1} x \rightarrow x = \cosh y$$

$$\theta = \tanh^{-1} \phi \rightarrow \theta = \tanh \phi$$

$$n = \coth^{-1} m \rightarrow n = \coth m$$

$$L = \operatorname{sech}^{-1} M \rightarrow \operatorname{sech} L = M$$

$$\alpha = \operatorname{cosech}^{-1} \beta \rightarrow \operatorname{cosech} \alpha = \beta$$

Example: Use a calculator to evaluate i) $\operatorname{ch} 4$ ii) $\operatorname{th} 2.3$

iii) $\operatorname{sech} 4.3$ v) $\operatorname{coth} 1.3$ vi) $\operatorname{cosech} 2$

solutions:

i) $\operatorname{ch} 4 = 27.31$

ii) $\operatorname{th} 2.3 = 0.98$

iii) $\operatorname{sech} 4.3 = 0.027$

v) $\operatorname{coth} 1.3 = 1.16$

vi) $\operatorname{cosech} 2 = 0.276$

Example: Use a calculator to evaluate

i) $\operatorname{sh}^{-1} 10$ ii) $\operatorname{th}^{-1} 0.4$

iii) $\operatorname{sech}^{-1} 0.5$ vi) $\operatorname{coth}^{-1} 8$

vii) $\operatorname{cosech}^{-1} 0.3$

Solution:

i) $\sinh^{-1} 10 = 2.99$

ii) $\tanh^{-1} 0.4 = 0.42$

iii) $\operatorname{sech}^{-1} 0.5 = \cosh^{-1} \frac{1}{0.5} = \cosh^{-1} 2$
 $= 1.316$

vi) $\coth^{-1} 8 = \tanh^{-1} \left(\frac{1}{8} \right) = 0.126$

vii) $\operatorname{cosech}^{-1} 0.3 = \sinh^{-1} \left(\frac{1}{0.3} \right) = 1.92$

Example: Solve for x

(a) $2 + 4 \sinh x = 3$

(b) $\frac{2 + 3 \tanh x}{e} = 10$

(c) $\ln (\operatorname{sech} x) = -0.5$

Solution:

(a) $2 + 4 \sinh x = 3 \Rightarrow \sinh x = \frac{3-2}{4}$

$$\sinh x = \frac{1}{4} \rightarrow x = \sinh^{-1} \frac{1}{4} = 0.247$$

(b) $e^{2+3 \tanh x} = 1$

$$\Rightarrow 2 + 3 \tanh x = \ln 1$$

$$2 + 3 \tanh x = 0$$

$$\Rightarrow \tanh x = -\frac{2}{3} \Rightarrow x = \tanh^{-1} \left(-\frac{2}{3} \right)$$
$$x = -0.805$$

(c) $\ln(\operatorname{sech} x) = -0.5$

$$\Rightarrow \operatorname{sech} x = e^{-0.5}$$

$$\Rightarrow x = \operatorname{sech}^{-1}(e^{-0.5})$$

$$= \cosh^{-1} \left(\frac{1}{e^{-0.5}} \right)$$

$$= \cosh^{-1}(e^{0.5})$$

$$= 1.085$$